American Computer Science League

2022-2023 ● Contest 3: Short Problems ● Junior Division

1. Boolean Algebra

Simplify this Boolean expression:

$$(A + B) + AB + B$$

A. B
B. A
C. AB
D. $A + B$
E. 1

2. Boolean Algebra

How many ordered pairs make the following Boolean expression TRUE?

$$(\overline{A}\, B)(A + \overline{B})$$

A. 0
B. 1
C. 2
D. 3
E. 4

3. Data Structures

Build the binary search tree for the following string:

K E N T U C K Y D E R B Y

List the nodes at depth 3 from left to right.

A. CKT
B. BDRU
C. CDTU
D. EY
E. Y

4. Data Structures

Given an initially empty stack and the following commands on the stack, what item will be popped next?

PUSH("M"), PUSH("E"), PUSH("R"), X=POP(), X=POP(), PUSH("C"), PUSH("H"), X=POP(), X=POP(), PUSH("A"), X=POP(), PUSH("N"), PUSH("T"), PUSH("O"), PUSH("F"), X=POP(), PUSH("V"), X=POP(), PUSH("E"), PUSH("N"), X=POP(), X=POP(), PUSH("I"), X=POP(), PUSH("C"), PUSH("E"), X=POP(), X=POP()

A. M
B. N
C. T
D. O
E. C

5. What Does This Program Do? (Arrays)

What is the output when the following program is run using the following values of array **val**?

4	7	2	6	7	3	8	3	5

```
n = 9
s = 0
for x = 0 to n-1
     s = s + val(x)
next x
a = s / n
t = 0
for x = 0 to n-1
     d = (val(x) - a) ^ 2
     t = t + d
next x
ans = sqrt(t / n)
output ans
```

A. 2
B. 4
C. 5
D. 9
E. 36